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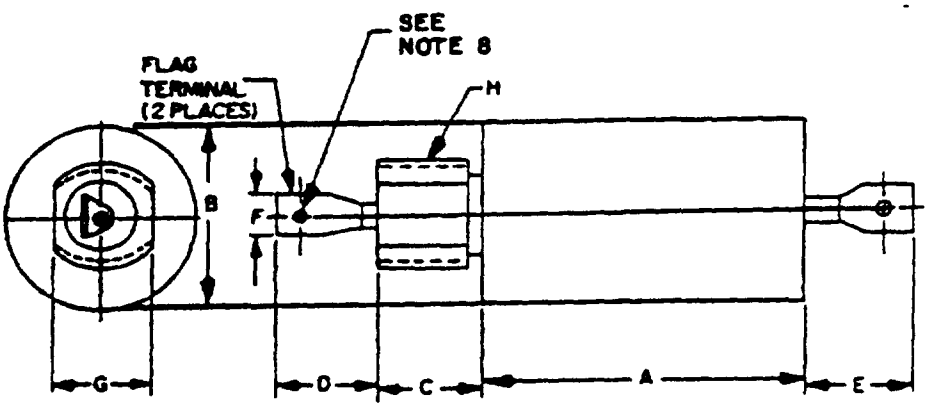
MIL-PRF-15733/54F
22 November 1983
SUPERSEDING
MIL-F-15733/54E(USAF)
8 March 1979

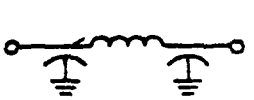
PERFORMANCE SPECIFICATION SHEET

FILTERS, RADIO FREQUENCY INTERFERENCE, HERMETICALLY SEALED, STYLE FL94

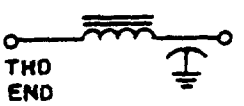
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the filters described herein shall consist of this specification and the latest issue of MIL-F-15733.





P1 (DASH 0001 THRU 0014)



L_2 (DASH 0015, 0018 AND 0019)



T (DASH 0017)

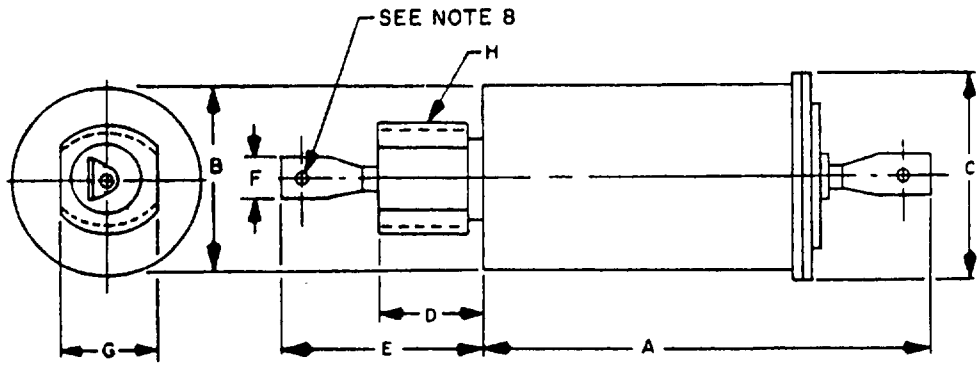
CIRCUIT CONFIGURATION

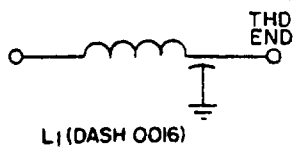
FIGURE 1. Case dimensions and circuit configurations.

(F) denotes changes

FSC 5915

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CIRCUIT CONFIGURATION

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.
4. Circuit diagram is for information only.
5. All filters shall be supplied with mounting hardware.
6. Case shall be tin-plated per MIL-T-10727. Terminals shall be hot tin dipped.
7. Use of style FL94 with or without shoulder is optional.
8. Wire hole diameter .060 (1.52 mm) (REF) for -0001 through -0016, -0018 and -0019, and .062 X .125 (1.57 X 3.18 mm) (REF) for -0017 (2 places).

FIGURE 1. Case dimensions and circuit configurations - Continued.

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Ⓕ TABLE I. Filter dimensions.

Dash number	A		B		C	D		E	F	G		H
	Min	Max	Min	Max	±.015	Min	Max	Max	Ref	Min	Max	Mounting thread
0001	.509 (12.93)	.571 (14.50)	.365 (9.27)	.385 (9.78)	.187 (4.75)	---	.150 (3.81)	.190 (4.83)	.115 (2.92)	.190 (4.83)	.210 (5.33)	.250-28 UNF-2A
0002	"	"	"	"	"	---	"	"	"	"	"	"
0003	"	"	"	"	"	---	"	"	"	"	"	"
0004	"	"	"	"	"	---	"	"	"	"	"	"
0005	"	"	"	"	"	---	"	"	"	"	"	"
0006	"	"	"	"	"	---	"	"	"	"	"	"
0007	"	"	"	"	"	---	"	"	"	"	"	"
0008	"	"	"	"	.312 (7.92)	---	"	"	"	"	"	"
0009	"	"	"	"	"	---	"	"	"	"	"	"
0010	"	"	"	"	"	---	"	"	"	"	"	"
0011	"	"	"	"	"	---	"	"	"	"	"	"
0012	"	"	"	"	"	---	"	"	"	"	"	"
0013	"	"	"	"	"	---	"	"	"	"	"	"
0014	"	"	"	"	"	---	"	"	"	"	"	"
0015	.680 (17.27)	.700 (17.78)	.344 (8.74)	.406 (10.31)	"	---	"	"	"	"	"	"
0016	1.149 (29.18)	1.179 (29.95)	.375 Typ. (9.52)	---	.410 (10.41)	.180 (4.57)	.200 (5.08)	.350 (8.89)	"	"	"	"
0017	---	.892 (22.66)	---	.680 (17.27)	.291 (7.39)	---	.285 (7.24) REF	.285 (7.24) REF	.125 (3.18) REF	.240 (6.10)	.260 (6.60)	.312-24 UNF-2A
0018	.530 (13.46)	.550 (13.97)	.365 (9.27)	.385 (9.78)	Max .197 (5.00) Min .177 (4.50)	---	.150 (3.81)	.190 (4.83)	.115 (2.92)	.190 (4.83)	.210 (5.33)	.250-28 UNF-2A
0019	.509 (12.93)	.571 (14.50)	.365 (9.27)	.385 (9.78)	.312 (7.92)	---	.150 (3.81)	.190 (4.83)	.115 (2.92)	.190 (4.83)	.210 (5.33)	.250-28 UNF-2A

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.
Case: Metal.
Terminals: Solderable (see figure 1).
Operating temperature range: -55°C to +125°C; temperature characteristic D.
Rated voltage: See table II.
Rated current: See table II.
DC resistance: See table II.
Insertion loss characteristic: Symbol W (see table II).
Creepage and clearance: Not applicable.
Seal: Method 112, MIL-STD-202, test condition A.
Capacitance to ground: Not applicable.
Temperature rise: 25°C, maximum.

Dielectric withstanding voltage: Method 301, MIL-STD-202.
Test temperature: 25°C.
Test voltage: 2.0 times rated dc voltage, applied between each terminal and ground, for a period of 1 to 5 seconds.
Maximum charging current: 50 mA.

Barometric pressure (reduced): Method 105, MIL-STD-202, test condition C (70,000 ft) for -0001 thru -0018 and test condition B (50,000 ft) for -0019.
Dielectric withstanding voltage: 125 percent of the rated voltage.
Maximum charging current: 50 mA.

Insulation resistance: Method 302, MIL-STD-202, test condition A.
Test temperature: 25°C.
Test potential: Rated dc voltage.
Maximum charging current: 50 mA.
Insulation resistance shall be not less than 100 megohms for -0001 thru -0016, -0018, and -0019 and not less than 1000 megohms for -0017.

Insertion loss: In accordance with MIL-STD-220 and table II.
Insertion loss (check test): In accordance with MIL-F-15733.

Overload: In accordance with MIL-F-15733.
Measurements at +25°C after test: Insulation resistance shall be measured and shall meet initial requirements.

Terminal strength: Method 211, MIL-STD-202, test condition A (pull).
Force: 5 pounds.

Resistance to soldering heat: Method 210, MIL-STD-202, test condition B.

Salt spray (corrosion): Method 101, MIL-STD-202, test condition A.

Thermal shock and immersion:
Thermal shock: Method 107, MIL-STD-202, test condition A, except the temperature at step 3 shall be +125°C.
Immersion: Method 104, MIL-STD-202, test condition A.
Measurements after final cycle:
Dielectric withstanding voltage with 90 percent of the initial voltage applied for 5 ±1 seconds. Maximum charging current: 50 mA.
Insulation resistance shall be not less than 30 percent of the initial requirement.
Insertion loss (check test) shall be measured and shall meet the initial requirements.

Shock (specified pulse): Method 213, MIL-STD-202, test condition A for -0001 thru -0017, and test condition I for -0018 and -0019.
Electrical load: 100 percent of the rated voltage between terminals and case.

Vibration, high frequency: Method 204, MIL-STD-202, test condition D (20 G) for -0001 thru -0014, -0016, -0017, and -0019; test condition B (15 G) for -0015 and -0018.
Electrical load: Rated voltage and current, as specified in table II, shall be applied continuously during test.

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Ⓕ TABLE II. Electrical characteristics.

Dash number	Circuit configuration	Maximum rated current (amperes)	Rated DC voltage (volts)	Rated AC voltage (volts) 0-400 Hz	Maximum DC resistance at 25°C (ohms)	Minimum full-load insertion loss (dB) in accordance with MIL-STD-220 at +25°C								
						.03 mHz	.05 mHz	.15 mHz	.3 mHz	.5 mHz	1 mHz	10 mHz	1 GHz	2 to 2.75 GHz
0001	Pi	10.0	300	115	0.003	--	--	14	20	--	30	70	70	--
0002	"	0.1	"	"	10	21	--	60	70	--	70	"	"	--
0003	"	0.25	"	"	4	8	--	50	66	--	70	"	"	--
0004	"	0.5	"	"	1	--	--	40	56	--	70	"	"	--
0005	"	1.0	"	"	0.25	--	--	23	43	--	70	"	"	--
0006	"	2.0	"	"	0.063	--	--	18	33	--	63	"	"	--
0007	"	3.0	"	"	0.027	--	--	14	25	--	54	"	"	--
0008	"	10.0	"	"	0.003	--	--	14	20	--	30	"	"	--
0009	"	0.1	"	"	10	21	--	60	70	--	70	"	"	--
0010	"	0.25	"	"	4	8	--	50	66	--	70	"	"	--
0011	"	0.5	"	"	1	--	--	40	56	--	70	"	"	--
0012	"	1.0	"	"	0.25	--	--	23	43	--	70	"	"	--
0013	"	2.0	"	"	0.063	--	--	18	33	--	63	"	"	--
0014	"	3.0	"	"	0.027	--	--	14	25	--	54	"	"	--
0015	L_2	1.0	200	"	0.25	2	5	18	26	--	46	"	"	--
0016	L_1	1.75	250	"	0.09	--	--	5	13	--	30	62	"	70
0017	T	10.0	300	"	0.006	5	--	20	23	27	35	60	60	--
0018	L_2	1.0	300	"	0.25	--	--	10	21	--	41	70	70	--
0019	L_2	10.0	300	"	0.003	--	--	8	14	--	25	45	70	--

Temperature conditioning (applicable to -0001 thru -0014 and -0016 only): Prior to vibration, filters shall be heated in an oven to +125°C plus one-half the temperature rise. Vibration in each plane shall begin within 5 minutes after removal of filters from the oven.
Test during vibration: During the last cycle in each direction, an electrical measurement shall be made to determine intermittent-open or short-circuiting.
After the vibration test, filters shall be subjected to and shall meet the initial requirements of the insertion loss (check test).

Moisture resistance: Method 107, MIL-STD-202, test condition A, except that the temperature at step 3 shall be +125°C. No measurement shall be made before and after cycling. Filters shall then be tested in accordance with method 106, MIL-STD-202.
Initial measurements: Not applicable.
Polarization voltage: 50 V dc.
Measurements after 24-hour drying period:
Dielectric withstanding voltage with 90 percent of the initial voltage applied for 5 ±1 seconds. Maximum charging current: 50 mA.
Insulation resistance shall be not less than 50 megohms.
Insertion loss (check test) shall be measured and shall meet the initial requirements.

Life: Method 108, MIL-STD-202.
Qualificaton inspection: Test condition D (1000 hours) for -0001 thru -0016, -0018, and -0019 and test condition B (250 hours) for -0017.
Group C inspection: Test condition B (250 hours).
Test temperature: +125°C.
Test voltage: 1.4 times the rated dc voltage.
Measurements after test:
Dielectric withstanding voltage with 90 percent of the initial voltage applied for 5 ±1 seconds. Maximum charging current: 50 mA.
Insulation resistance shall be not less than 30 percent of the initial requirement.
Insertion loss (check test) shall be measured and shall meet the initial requirements.

Part number: M15733/54 - (dash number from table I).

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Navy - SH
Air Force - 11, 17, 99
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Navy - AS, MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-15733/54F	FILTERS, RADIO FREQUENCY INTERFERENCE, HERMETICALLY SEALED, STYLE FL94

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
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